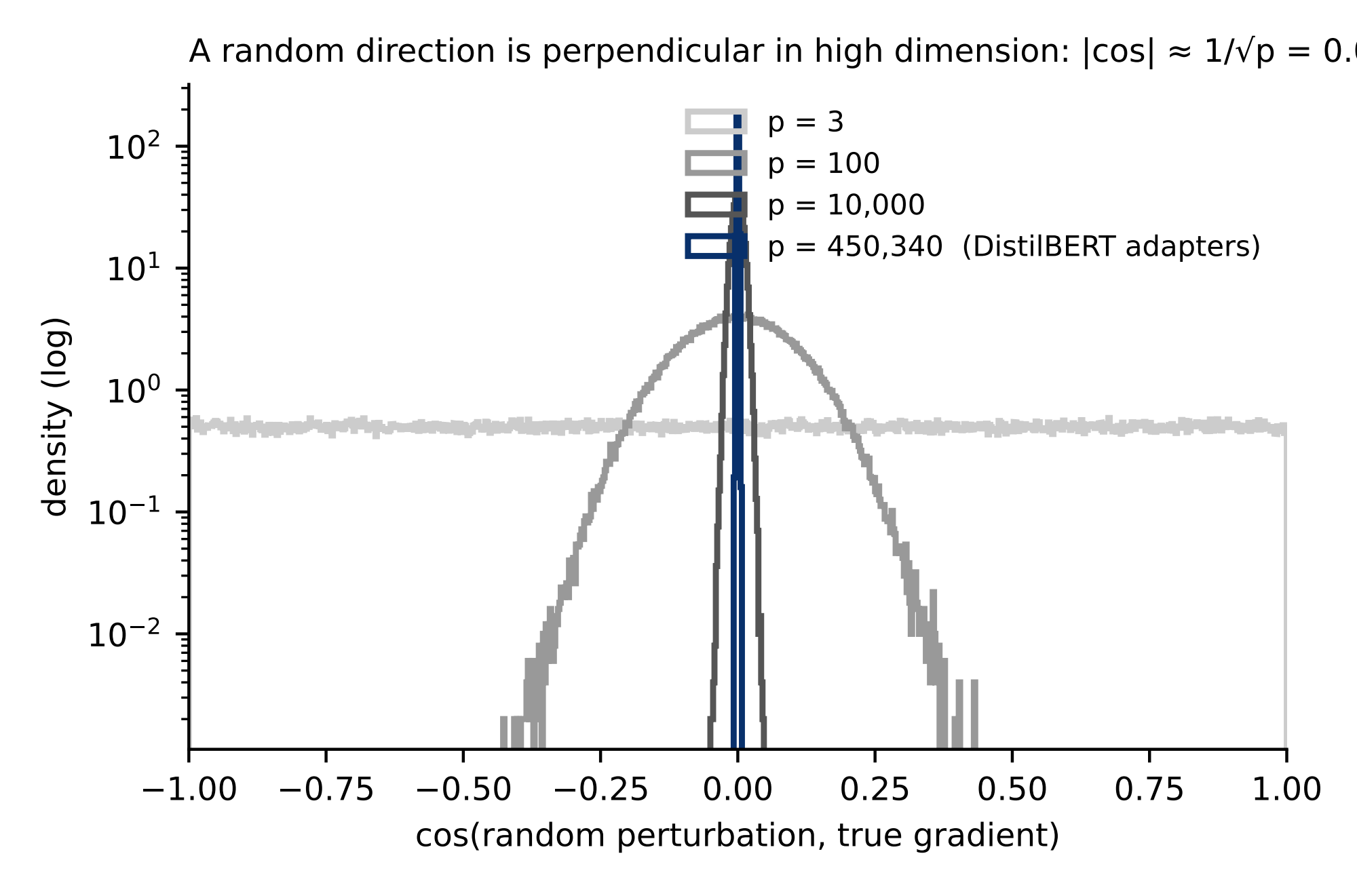
A random direction is perpendicular in high dimension: |cos| ≈ 1/√p = 0.
p = 3
p = 100
p = 10,000
p = 450,340 (DistilBERT adapters)
density (log)
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-2}
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00
cos(random perturbation, true gradient)